

AKSHAY CHAUHAN

Legally Authorized to Work in Canada (Permanent Resident)
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PROFESSIONAL SUMMARY

Data Analyst with hands on experience delivering client facing analytics, building ETL pipelines, designing dimensional data warehouses and developing interactive Tableau/Power BI dashboards using Python, SQL, PostgreSQL and Excel. Skilled in transforming raw, disparate data into clear, actionable business insights through data cleaning, modeling and KPI reporting. Brings cross industry experience (retail, manufacturing, healthcare and consulting) with strong communication skills to translate ambiguous business questions into data driven recommendations stakeholders can act on.

TECHNICAL SKILLS

Programming Languages: Python, SQL

Data Analysis & Visualization: Power BI, Tableau, Microsoft Excel (Pivot Tables, Formulas, VLOOKUP/XLOOKUP, Power Query), Pandas, NumPy

Databases: PostgreSQL, DuckDB, SQLAlchemy, Relational Databases

Data Engineering: ETL, Data Pipelines, Real Time Data Ingestion, Data Cleaning, Data Transformation, Deduplication, Dimensional Modeling, Star Schema Design, Data Warehousing, Data Lakes (Parquet, Date Partitioning), Protocol Buffers (Protobuf), GTFS-Realtime, Data Validation / Data Quality Assurance

Tools & Technologies: Git, GitHub, Jupyter Notebook, VS Code, FastAPI, REST APIs, Groq API, pytest, unittest, Salesforce CRM, ServiceNow, Supabase

Core Concepts: Business Intelligence, Data Visualization, Dashboard Development, KPI Reporting, Agile/Scrum, LLM Integration, Text-to-SQL, SQL Guardrails

PERSONAL PROJECTS / PORTFOLIO WORK

DuckStream: Live Transit Data Pipeline

2026

[GitHub](#) | Python, DuckDB, Parquet, FastAPI, GTFS-Realtime, Groq

- Built an unattended ingestion pipeline polling the OC Transpo GTFS-Realtime feed every 45 seconds, decoding 83.1M protobuf records at 99.9% poll coverage, replayed from a crash-safe raw archive into a date-partitioned Parquet lake of 50.8M deduplicated rows (720 MB to 200 MB, 3.6x).
- Cut 39% of rows as redundant by measuring the feed's real report interval (median 60 seconds) instead of assuming it, and proved (vehicle_id, timestamp) safe as a dedupe key across 417,105 duplicates with zero conflicts, which also corrected the polling rate from 30 to 45 seconds.
- Caught a silent data quality failure where 31% of feed predictions are the published timetable echoed back verbatim, which had reported the median delay as 0 seconds; excluding those rows moved it to 90 seconds.
- Derived true arrival delay by joining 41.9M predictions to measured-match timetable versions (97-99% trip coverage, up from 31%), producing 585,201 measured arrivals across 5,544 stops and 177 routes: median bus 167 seconds late, only 23% within a minute of schedule, rising to a 225 second median at 4-6 pm.
- Shipped a public FastAPI demo translating plain English questions into DuckDB SQL with an LLM, checked by a guardrail before execution: single SELECT only, read-only connection, external file access disabled, row limit and a 10 second watchdog. 128 tests.

Retail Analytics Data Warehouse and Business Intelligence Dashboard

2026

[GitHub](#) | Python, Pandas, SQL, PostgreSQL, Power BI, Jupyter Notebook

- Built an ETL pipeline using Python, Pandas and SQL to extract, clean and load raw multi table retail data into PostgreSQL.
- Designed a star schema data model with fact and dimension tables to support efficient, scalable analytical querying.
- Wrote SQL queries to analyze sales trends, customer behavior and product and seller performance across the warehouse.
- Built interactive Power BI dashboards visualizing key business KPIs, including sales trends, category performance and customer insights.
- Validated and cleaned data throughout the pipeline to ensure accuracy and consistency across all reporting layers.

Clafy: Know What You Eat, Android Mobile Application

2026

[Google Play Store](#) | React Native, Expo, Supabase, PostgreSQL, REST APIs

- Developed and published a production ready Android application on Google Play, taking it from concept through design, development, testing and release.
- Designed a scalable PostgreSQL backend on Supabase to manage user authentication, personalized health profiles, scan history and product data.
- Integrated the Open Food Facts API to retrieve real time nutritional data from a global database of millions of products.
- Engineered a centralized ingredient matching engine supporting 400+ ingredients with 1,000+ aliases and E number mappings to power accurate, personalized health recommendations.
- Architected a persistent caching layer that stores newly scanned products in PostgreSQL, reducing external API dependency and improving lookup performance on repeat scans.

- Performed end to end testing and debugging across authentication, API integrations and database operations prior to production deployment.

Codeforces Performance Analyzer and AI Coach

2026

[GitHub](#) | Python, REST APIs, Groq API, Llama 3.3, Data Analysis

- Integrated the Codeforces REST API to retrieve user profiles, contest history and submission data for performance analysis.
- Built a metrics pipeline calculating acceptance rate, problems solved, average problem difficulty, contest participation and activity streaks.
- Developed a tag based failure rate algorithm to identify a user's weakest problem solving areas, enabling targeted, personalized recommendations.
- Structured analytical output into prompts for a Large Language Model (Llama 3.3 via Groq API) to generate a personalized performance review and a 30 day improvement roadmap.
- Designed a modular architecture separating data collection, analytics, prompt generation and AI integration for maintainability and scalability.
- Implemented error handling and input validation across API calls to ensure reliable data retrieval and report generation.

PROFESSIONAL EXPERIENCE

Data Analyst, Jash Data Science, Remote

Jan 2024 to Sept 2026

- Partnered with client teams to define KPIs and reporting requirements, translating ambiguous business questions into structured analytical frameworks.
- Built and maintained interactive Tableau/Power BI dashboards tracking key performance metrics, used by client leadership for weekly decision making.
- Wrote complex SQL queries to extract, clean and join data from multiple disparate client systems (CRM, ERP, legacy databases) into unified analysis ready datasets.
- Automated recurring client reporting workflows using Python (pandas), cutting manual reporting time by 10+ hours per week.
- Drove a 15% reduction in marketing campaign costs and a 40% increase in potential customer reach by analyzing audience data and reallocating budget toward higher performing channels.
- Contributed to proposal and pitch materials by conducting preliminary data analysis to support new business development.

Data Analyst (Quality Control), Maple Leaf, Winnipeg, MB

Jan 2023 to Dec 2023

- Built and maintained Excel based reporting models (pivot tables, formulas, trend analysis) to track quality metrics across construction material product lines, delivering findings directly to the quality manager.
- Analyzed production and inspection data to identify recurring process gaps, several of which led to corrective actions that noticeably reduced repeat defects on the affected lines.
- Delivered weekly quality reporting to leadership, translating raw inspection data into clear, decision ready summaries for stakeholders.
- Investigated client complaints using historical quality data to diagnose root causes, closing issues with data backed corrective actions rather than one off fixes.

Technology Support Analyst, Region of Peel, Toronto, ON (Contract)

Jan 2022 to Jan 2023

- Resolved 50 to 100 weekly support tickets via ServiceNow, tracking recurring issue patterns to prioritize fixes and reduce future ticket volume.
- Reduced repeat ticket volume by an estimated 15 to 20% by identifying issue trends and feeding them into structured knowledge base documentation.
- Documented recurring procedures and trends, contributing data that supported broader reporting on IT support performance.

IT Support Specialist, Trillium Health Partners, Toronto, ON (Contract)

Jun 2021 to Dec 2021

- Validated and corrected patient and user records during an Epic EMR system migration, identifying data discrepancies before go live to protect record accuracy in a healthcare compliance environment.
- Tracked and categorized recurring user issues during the migration, feeding that pattern data into a streamlined support knowledge base for the wider team.
- Triaged help desk tickets by urgency, prioritizing cases with direct patient care impact over routine requests.

EDUCATION

Computer Programming, Algonquin College, Ottawa

Jan 2018 to Dec 2020